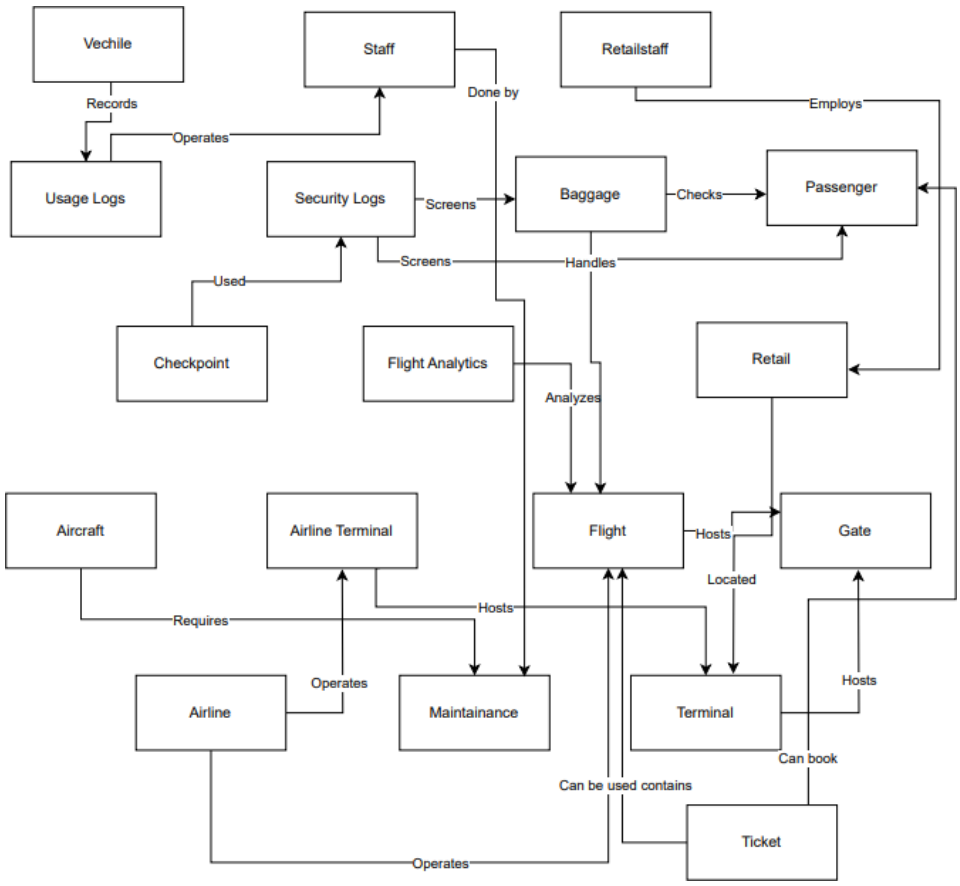
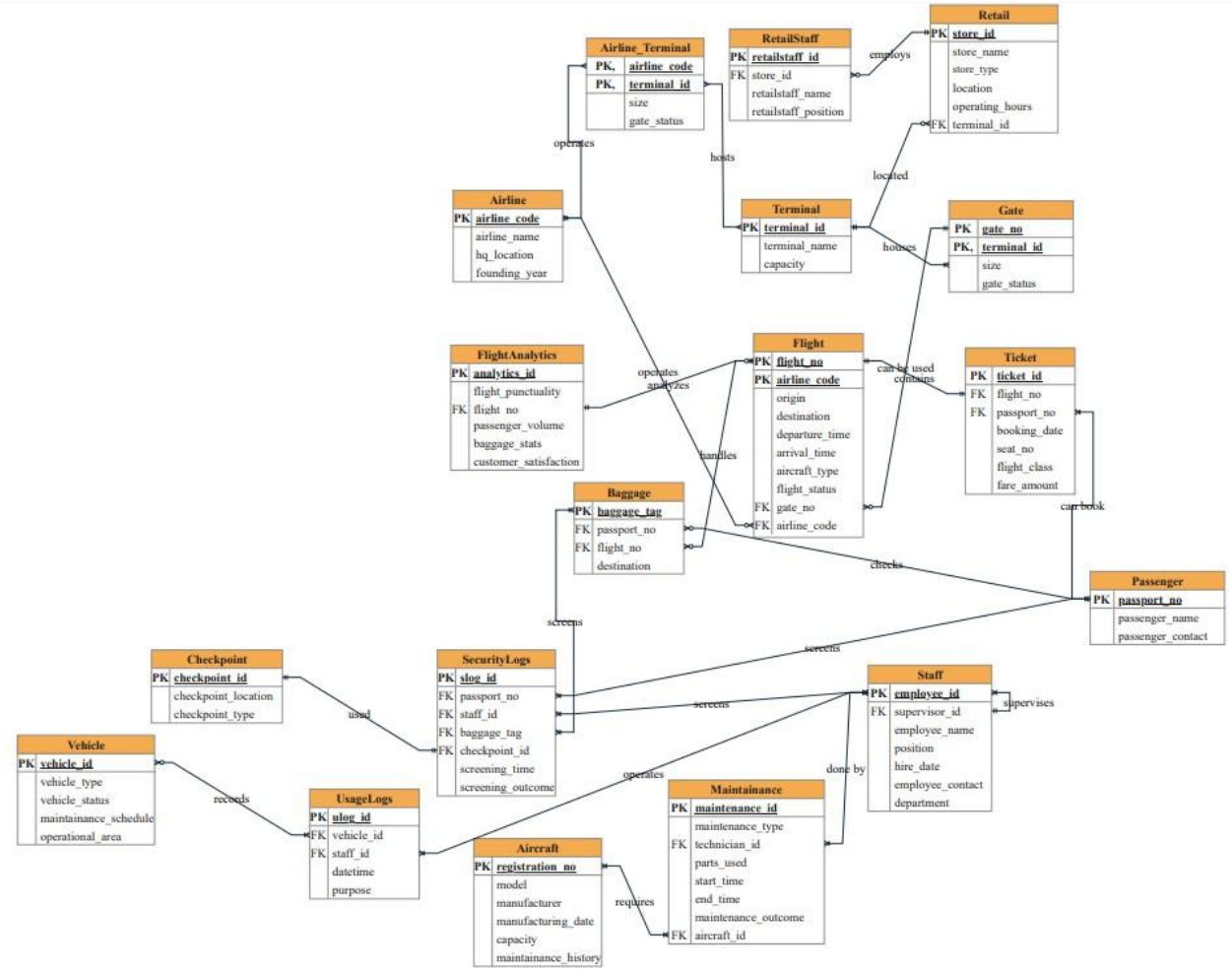
Assignment 1: Airport Management System

Group 2 (4 members)

1. Entity Relationship Diagram:





2. Relationships in both directions:

- **Staff – Staff**
    - Staff supervises staff
    - Staff is supervises by staff
- **Terminal – Gate**
    - Terminal houses gate
    - Gate is houses by terminal
- **Airline – Flight**
    - Airline operates flight
    - Flight is operates by airline
- **Terminal – Retail**

- Terminal contains retail
- Retail is contains by terminal

- **Flight – Gate**
  - Flight uses gate
  - Gate is uses by flight
- **Retail – Terminal**
  - Retail located terminal
  - Terminal is located by retail
- **Retail – Staff**
  - Retail employs staff
  - Staff is employs by retail
- **Flight – Baggage**
  - Flight handles baggage
  - Baggage is handles by flight
- **Passenger – Baggage**
  - Passenger checks baggage
  - Baggage is checks by passenger
- **Checkpoint – Passenger**
  - Checkpoint screens passenger
  - Passenger is screens by checkpoint
- **Flight – Gate**
  - Flight assigned gate
  - Gate is assigned by flight
- **Airline – Terminal**
  - Airline assigned terminal
  - Terminal is assigned by airline
- **Ticket – Passenger**
  - Ticket booked passenger
  - Passenger is booked by ticket

- **Staff – Department**
  - Staff serves department
  - Department is serves by staff

- **Retail – Terminal**
    - Retail leases terminal
    - Terminal is leases by retail
- **Vehicle – Staff**
    - Vehicle drives staff
    - Staff is drives by vehicle
- **Aircraft – Maintenance**
    - Aircraft maintains maintenance
    - Maintenance is maintains by aircraft
- **Maintenance – History**
    - Maintenance records history
    - History is records by maintenance
- **Flight – History**
    - Flight linked history
    - History is linked by flight
- **Flight – Passenger**
    - Flight boarded passenger
    - Passenger is boarded by flight
- **Flight – Analytics**
    - Flight associated analytics
    - Analytics is associated by flight

3. Data Dictionary

Aircraft

Field (PK/FK)	Data Type	Description
registration_no (PK)	VARCHAR	Aircraft registration number
model	VARCHAR	Aircraft model
manufacturer	VARCHAR	Manufacturer
manufacturing_date	DATE	Date of manufacturing
capacity	INT	Passenger capacity
maintainance_history	TEXT	Maintenance history

Airline

Field (PK/FK)	Data Type	Description
airline_code (PK)	VARCHAR	Unique code of the airline
airline_name	VARCHAR	Name of the airline
hq_location	VARCHAR	Headquarters location
founding_year	YEAR	Year of establishment

Baggage

Field (PK/FK)	Data Type	Description
baggage_tag (PK)	VARCHAR	Unique baggage identifier
passport_no (FK)	VARCHAR	Linked passenger
flight_no (FK)	VARCHAR	Linked flight
destination	VARCHAR	Baggage destination

Checkpoint

Field (PK/FK)	Data Type	Description
checkpoint_id (PK)	INT	Checkpoint ID
checkpoint_location	VARCHAR	Location of checkpoint
checkpoint_type	VARCHAR	Type of checkpoint

Flight

Field (PK/FK)	Data Type	Description
flight_no (PK)	VARCHAR	Flight number
airline_code (PK, FK)	VARCHAR	Associated airline
origin	VARCHAR	Origin airport
destination	VARCHAR	Destination airport
departure_time	DATETIME	Scheduled departure time
arrival_time	DATETIME	Scheduled arrival time
aircraft_type	VARCHAR	Aircraft model/type
flight_status	VARCHAR	Status (on time, delayed, etc.)
gate_no (FK)	VARCHAR	Associated gate number

FlightAnalytics

Field (PK/FK)	Data Type	Description
analytics_id (PK)	INT	Analytics record ID
flight_no (FK)	VARCHAR	Related flight
flight_punctuality	VARCHAR	Timeliness of flight
passenger_volume	INT	Passenger count
baggage_stats	TEXT	Baggage-related stats
customer_satisfaction	VARCHAR	Feedback

Gate

Field (PK/FK)	Data Type	Description
gate_no (PK)	VARCHAR	Gate number within terminal
terminal_id (PK, FK)	INT	Associated terminal ID
size	VARCHAR	Size classification (small/medium/large)
gate_status	VARCHAR	Status of the gate

Maintainance

Field (PK/FK)	Data Type	Description
maintenance_id (PK)	INT	Maintenance record ID
maintenance_type	VARCHAR	Type of maintenance
technician_id (FK)	INT	Staff performing maintenance
parts_used	TEXT	Parts used in maintenance
start_time	DATETIME	Start time
end_time	DATETIME	End time
maintenance_outcome	VARCHAR	Maintenance result
aircraft_id (FK)	VARCHAR	Aircraft maintained

Passenger

Field (PK/FK)	Data Type	Description
passport_no (PK)	VARCHAR	Unique identifier for a passenger
passenger_name	VARCHAR	Name of the passenger
passenger_contact	VARCHAR	Contact details

Retail

Field (PK/FK)	Data Type	Description
store_id (PK)	INT	Retail store identifier
store_name	VARCHAR	Name of the store
store_type	VARCHAR	Type of store
location	VARCHAR	Store location in terminal
operating_hours	VARCHAR	Business hours
terminal_id (FK)	INT	Linked terminal

RetailStaff

Field (PK/FK)	Data Type	Description
retailstaff_id (PK)	INT	Staff ID
store_id (FK)	INT	Assigned store
retailstaff_name	VARCHAR	Staff name
retailstaff_position	VARCHAR	Staff role/position

SecurityLogs

Field (PK/FK)	Data Type	Description
slog_id (PK)	INT	Security log identifier
passport_no (FK)	VARCHAR	Involved passenger
staff_id (FK)	INT	Screening staff
baggage_tag (FK)	VARCHAR	Baggage involved
checkpoint_id (FK)	INT	Security checkpoint
screening_time	DATETIME	Time of screening
screening_outcome	VARCHAR	Result of screening

Staff

Field (PK/FK)	Data Type	Description
employee_id (PK)	INT	Staff unique identifier
supervisor_id (FK)	INT	Reports to supervisor
employee_name	VARCHAR	Name of the staff
position	VARCHAR	Job title
hire_date	DATE	Date of joining
employee_contact	VARCHAR	Contact details
department	VARCHAR	Department assigned

Terminal

Field (PK/FK)	Data Type	Description
terminal_id (PK)	INT	Unique identifier for the terminal
terminal_name	VARCHAR	Name of the terminal
capacity	INT	Passenger capacity of the terminal

Ticket

Field (PK/FK)	Data Type	Description
ticket_id (PK)	VARCHAR	Unique ticket identifier
flight_no (FK)	VARCHAR	Linked flight number
passport_no (FK)	VARCHAR	Linked passenger passport number
booking_date	DATE	Date of booking
seat_no	VARCHAR	Assigned seat number
flight_class	VARCHAR	Class of service
fare_amount	DECIMAL	Fare paid for the ticket

UsageLogs

Field (PK/FK)	Data Type	Description
ulog_id (PK)	INT	Usage log ID
vehicle_id (FK)	INT	Vehicle used
staff_id (FK)	INT	Staff using the vehicle
datetime	DATETIME	Date and time of usage
purpose	VARCHAR	Purpose of usage

Vehicle

Field (PK/FK)	Data Type	Description
vehicle_id (PK)	INT	Vehicle identifier
vehicle_type	VARCHAR	Type of vehicle
vehicle_status	VARCHAR	Status of the vehicle
maintainance_schedule	VARCHAR	Scheduled maintenance
operational_area	VARCHAR	Operating zone